

Connor Patrick Hyde

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EDUCATION

Cornell University - Duffield College of Engineering
Bachelor of Science in Mechanical Engineering | GPA: 3.66

Ithaca, NY
Expected May 2028

Relevant Coursework: Fluid Mechanics, System Dynamics, Mechanics of Engineering Materials, Dynamics, Statics & Mechanics of Solids, Thermodynamics, Introduction to Mechanical Design

PROJECTS

Frictional Braking System - [View Project](#) October 2025 - March 2026

- Designed brakes in **SolidWorks** utilizing a C-bracket and vertical rails to achieve **dual-clamping** on I-beam flange
- Reduced subsystem size by **27%** and increased braking force **10x** by pairing stronger springs with a larger actuator
- Developed **MATLAB** script mapping spring constant vs. free length across different design constraints to identify springs that provide enough braking force to reliably stop a **100 kg** pod in less than **half a second**
- Validated design in **Ansys** achieving a factor of safety of **>3.0** with peak deflection under **0.02 in** at expected loads
- Machined **12** precision parts on manual lathe and mill working within tolerances of **+/- 0.005in**
- Eliminated **3** redundant components from pneumatics circuit, and substituted the existing regulator capable of **3000 psi** inlet pressure, enabling hundreds more braking events per tank refill
- Reduced operating pressure by **~50%** to **65psi** to decrease component wear, chances of leaks, and improve safety

Open Arm Hardware Casing - [View Project](#) July 2026 - August 2026

- Designed OpenArm exterior casing in **Onshape** on a 2-member team, maximizing hardware and electrical coverage
- Enhanced heat dissipation through ventilation slots and improved wire management via integrated cable hooks
- Created **MATLAB** GUI to model arm kinematics and calculate peak end-effector velocity; used results to define casing load requirements and prioritize rapid replaceability over wall thickness
- Optimized bed layout to cut total print time down by **44%**, and maximized yield strength across all **30** covers by aligning layer deposition direction with primary load paths and tuning infill and wall count

EXPERIENCE & INTERNSHIPS

Cornell Hyperloop - Track Team Lead & Machining Lead Ithaca, NY | October 2024 - Present

- Developing advanced track geometry to allow top-down levitation, amplify propulsion, and strengthen guidance
- Coordinating machining for a **20-person** mechanical team producing **60+ parts**, tracking tasks in Excel, assigning fabrication with deadlines, and reserving machines weekly to ensure on-time system integration
- Presented magnetic levitation and socioeconomic research to a panel of competition judges at **2026 European Hyperloop Week** in Veendam, Netherlands. Received **5th place** in division for overall pod design at EHW
- Tested and quantified regenerative braking performance using an actuated motor wheel on a rotating flywheel

Robotics Center of Silicon Valley (YC '25) - Robotics Engineer San Francisco, CA | June 2026 - August 2026

- Represented the company at open houses and private events, demonstrating Robotics Center's mobile manipulator and Aloha Arm teleoperation capabilities to **450** attending clients, investors, and industry peers
- Worked closely with electrical and business teams to guarantee on-time preparation and fulfillment of orders
- Inspected shipments and resolved assembly errors prior to **5** OpenArm shipments to ensure customer satisfaction
- Diagnosed and resolved recurring print-quality and operational issues across **4** Bambu Lab 3D printers

Autodesk Design and Make Space - Lead Machining Instructor Ithaca, NY | August 2025 - Present

- Certified as a **green apron machinist** with **125+ hours** of shop experience, authorized to operate advanced programmable machinery and proficient in manual lathe and mill operations
- Delivered **75+** hours of machining instruction through **2.5-hour** training sessions, qualifying users for red aprons by teaching shop safety, and core lathe and mill operations including facing, drilling, and tapping

SKILLS

Software and Programming: Ansys, SolidWorks, Onshape, Fusion, MATLAB, Simulink, Python, Java, Excel